(*d*) For single-phase half bridge inverter the input voltage is 200 V and feeds resistive load of 5 Ω. Determine : 7

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(*i*) RMS output voltage

(*ii*) The average current of each power device

(*iii*) The output power P_o

(*iv*) 3rd and 5th harmonic rms current at output.

5. (*a*) Write four applications of cycloconverter. 2

(*b*) Explain the operation of single-phase to single-phase cycloconverter feeding RL load. Draw appropriate waveforms. 7

(*c*) Discuss the principle of phase control in single-phase full-wave ac voltage controller. Derive expression for the rms value of its output voltage. 7

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(*d*) A single-phase AC voltage controller is supplied with 230 V ac voltage. If resistive load is 1 kW rating at 230 V, rating voltage is connected to AC voltage controller. Calculate the firing angle if the load to consume 50% power of its rated power at rated voltage. 7

325654(25)

BE (6th Semester) Examination, April-May, 2016

(New Scheme)

Power Electronics Devices & Circuits

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Time Allowed : 3 hours *Maximum Marks* : 80

Minimum Pass Marks : 28

Note : (*i*) Part (*a*) is compulsory. Solve any **two** parts from (*b*), (*c*) and (*d*) of each question.

(*ii*) The figures in the right-hand margin indicate marks.

1. (*a*) Justify the statement "Gate signal is not necessary and is also not preferred after the SCR starts conducting". 2

(*b*) Describe the dynamic characteristic of SCR with neat sketches. 7

(*c*) The gate-cathode characteristic of SCR is given by $V_g = 0.5 + 8\ I_g$. For a triggering frequency of 400 Hz and duty cycle of 0.1, compute the on period of the pulse and value of resistance to be connected in series with gate circuit. The gate pulse applied has amplitude of 12 V. Average gate power dissipation of SCR is 0.5 W. 7

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(*d*) An SCR of current rating 100 A is to be used in parallel with 200 A SCR. Their on-state drops are 2 V and 1.4 V respectively. Calculate the series resistance that should be connected in series with each SCR if two SCRs have to share total current of 250 A in porportion to their ratings. 7

2. (*a*) Justify the statement "use of freewheeling diode increases the efficiency and system power factor in converter circuit in R-L load condition. 2

(*b*) Draw neat sketches of the output voltage and current waveforms of a single-phase fully controlled bridge converter without freewheeling diode feeding an RLE load with extinction angle β less than firing angle α. 7

(*c*) What is the effect of source inductance on the operation of single-phase full bridge converters ? Derive expression for output voltage, including the effect of source inductance. 7

(*d*) A six-pulse thyristor converter connected on the secondary of a delta star connected 6.6 kV / 415 V, 50 Hz transformer supply 460 V, 200 A dc load, Find : 7

(*i*) Converter firing angle

(*ii*) DC power delivered by converter

(*iii*) AC line current

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(*iv*) RMS valve of thyristor current

(*v*) Average thyristor current

3. (*a*) Define the duty cycle related to chopper. Also name the two control strategies to control the duty cycle. 2

(*b*) What is the need of commutation in thyristor circuits ? List the different methods of commutation schemes and explain with relevant waveforms the working of resonant-pulse commutation technique. 7

(*c*) Explain the working of a current commutated chopper with the help of voltage and current waveforms. 7

(*d*) A step-up chopper has output voltage is 2 to 4 times the input voltage for a chopper frequency of 2000 Hz. Determine the range of OFF time period of the gate signal. 7

4. (*a*) What are the disadvantages of the harmonics present in the inverter system ? 2

(*b*) Discuss the principle of working of a three-phase bridge inverter with an appropriate circuit diagram. Draw phase and line voltage waveforms on the assumption that each thyristor conducts for 180° and the resistive load is star connected. 7

(*c*) What is pulse width modulation technique? List different PWM techniques and explain with relevant diagram the sinusoidal pulse width modulation technique. 7

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